AI for Good Sandbox Hackathon: Technical Report

Team [Team name]: SarkiAI Policy Sandbox (Cameroon)

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Github: https://github.com/rouftom/sarkiai-policy-sandbox
Domain: public governance & health data
Alignment: ITU-T Y.3172 · ITU Readiness 2.0 (6 factors, 13 dimensions)

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1. Introduction

AI is increasingly used to analyse anomalies, but detecting one is only half the problem. The other half (**how institutions respond**, who is accountable, under which mandate, within which deadline) is where public systems fail: slowly, ad hoc, and with uncertain legal footing.

We propose **SarkiAI Policy Sandbox**: a policy-grounded multi-agent engine that ingests a country's legal corpus into a jurisdiction-scoped knowledge base, runs **mandated institutional agents** with partial information, executes **progressive error scenarios** in sequenced blocks, and produces **auditable** joint reports with a hash-chained log and persisted traces. It scores the **13 ITU readiness dimensions** on 0–5, projects causal what-if effects, and carries scanned gazette pages through OCR into an impact analysis on open incidents.

Sovereignty is architectural, not a setting. Generation (`qwen3.5:9b`) and embeddings (`bge-m3`, 1024-d) run on-premise in Ollama; no request leaves the host, and there is no cloud fallback to disable.

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2. Description of the use case

The case. Institutional response to a **health-data breach** in a public hospital connected to a national e-health platform. Then, in the same engine and corpus, the **exposure of minors' records** on a national learning platform.

The problem. Detection is mature; the response is not. Notification duties are split across the health ministry, the data-protection authority and the cyber agency. No actor holds the full picture, the law is silent on several hand-offs, and the outcome depends on who read which document first.

Gaps in existing solutions. Notification duties are not allocated across sectors, so each authority waits for another. Decisions are not machine-readably audited, so accountability cannot be reconstructed. The tension between the duty to notify and the duty to preserve continuity of service is unresolved by any instrument we found. Generic assistants answer with fluent prose and no traceable basis, and cannot refuse cleanly when the corpus is silent. Cloud inference on this material is a sovereignty risk we consider unacceptable.

How the system answers. Agents are grounded in national provisions with **article-level citations verified against the retrieved text**: an invented quote is rejected and generation retried with a directive naming the passage. Mandate conflicts are surfaced, not smoothed over. Every step lands in a tamper-evident audit trail and a persisted trace, exportable as a run dossier. Local inference removes the sovereignty risk by construction, not by policy.

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3. Mapping with ITU-T Y.3172 and Readiness 2.0

Each Y.3172 node maps onto a component of the deployment, carries a requirement, and is scored against the Readiness 2.0 dimensions it serves.

Node	Implemented as · requirement	Dimension(s)
SRC	logs, EHR extracts, telemetry, gazettes · coverage, retention, access control	1 Marketplace · 13 Infrastructure
C	chunker (FR/EN), OCR, transcription · fidelity to the published text; no pseudo-law from a bad scan	5 Workflows · 13 Infrastructure
PP	normalisation, jurisdiction tagging, de-identification · purpose limitation, per-chunk provenance	1 Marketplace · 6 Human Interface
M	local generation + hybrid retrieval (bge-m3 + BM25, RRF, optional rerank) · no external inference	5 Workflows · 3 Cross-Domain Correlation
P	citation gate, routing, approvals, scoring · a legal basis per claim; human validation before any side effect	10 AI & Policies · 7 Strategy Alignment
D	API, console, signed approval links, console outbox · alignment across institutions	7 Strategy Alignment · 8 Collaboration
SINK	agents, officials, joint report, dossier · oversight, inclusion	9 Impacts of Humans · 6 Human Interface · 11 Inclusion

Anchoring documents: the 2024 data-protection law and the CEMAC directive (PP, P), cybersecurity law 2010/012 (C, P), the communications law and the ART compendium (SRC), SND30 and PSNSN (D, SINK). **Sovereignty cuts across all nodes**: generation, embeddings and reranking run on the deployment host, so patient data never leaves national infrastructure (dimensions 10 and 11). The compose file contains no external inference service, and the provider configuration exposes one mode, local.

3.1 Knowledge base

Cameroon: 26 manifest entries, 21 ingestible today: the 2024 data-protection law, cybersecurity law 2010/012, communications and e-commerce laws, penal code, the MINPOSTEL compendium 2023, the ART compendium 2018 (486 pp.), SND30, the national digital strategy, the digital-health plan (PSNSN 2026-2030), two decrees and the CEMAC users-rights directive. Five entries have no text: the constitution, the 1996 framework law, the DPA and ANTIC mandates, and the cyber strategy, all to_confirm. They stay listed, not dropped.

Each chunk carries country, jurisdiction, docType, pillar, headingPath and articleNumber. Eight documents carry an OCR caveat (six with no text layer, one partial, one non-Unicode font) and go through local Tesseract with a zone overlay, so an operator sees the exact crop the text came from.

Ten regional instruments are ingested for comparison: AU Continental AI Strategy 2024, Kigali Declaration 2025, Windhoek Statement, Smart Africa blueprint, and the strategies of Kenya (with roadmap), Rwanda, Egypt, Zambia and Nigeria. **Six international references are catalogued but not loaded**: ITU-T Y.3172, the ITU Readiness Framework, WHO, UNESCO, Malabo, OECD: so the system cannot quote them; they frame the gaps, nothing more. The ITU taxonomy (6 factors, 13 dimensions, 111 metrics) is shipped as data/global/itu/dimensions.json and mirrored in the scoring engine.

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4. What is deterministic, and what is the model

Step	Mechanism	Arguable?
Retrieval	Qdrant dense + BM25, RRF, optional rerank	no: arithmetic over vectors
Citation extraction	quotes matched against the retrieved corpus	no: exact substring match
Citation gate	validateCitations: absent quote → rejection, retry with a named directive	no: a loop, not a suggestion
Readiness score	weighted composite; causal propagation over a fixed graph	no: arithmetic
Scenario	tick release, per-agent visibility sets	no: declared in the record

Step	Mechanism	Arguable?
sequencing		
Audit	hash chain, end-to-end re-verification	no: cryptographic
Approval	signed token, single use, expiry, bound to one action	no: signature check
Drafting, deciding when to search, ranking impact	qwen3.5:9b, on-demand retrieval	**yes**: hence gated, cited, logged

The model writes; the system decides what may be written. The console shows rejected quotes when a retry happened and the audit records the attempt count, so the gate can be observed rather than trusted.

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5. Error scenarios

Facts are released in blocks; each institution sees only its block, then the coordinator reconciles the independent assessments.

A: health-data breach. Tick 0: anomalous access on the e-health platform (ministry, cyber agency). Tick 1: possible exfiltration of patient records (+ DPA). Tick 2: media pressure and a notification deadline. Output: the **mandate conflict** between the DPA's notification duty and the ministry's continuity-of-care duty, the open questions, and the gaps: each with its article, or an explicit statement that none was found.

B: minors' records, same corpus, different domain. Tick 0: a school reports that the learning platform shows pupils from other schools. Tick 1: ~40,000 pupil records (names, birth dates, results) reachable for days, some pupils under 13. Tick 2: parents demand notification within 72 hours; the ministry must decide whether to suspend the platform. Awkward on purpose: two ministries and a regulator share the decision, the subjects are minors, and the service cannot simply be switched off. The engine must name the provisions that apply and say plainly where the corpus is silent, instead of inventing a 72-hour rule.

C: a gazette that changes open cases. A scanned page of the Official Gazette is analysed zone by zone, OCR'd and ingested as one document. The panel asks what the new provision changes for open runs: severity, deadlines, runs untouched. The provision is analysed, not trusted; the zones stay visible so a wrong cut is caught before the text becomes retrievable.

When the corpus is silent, we say so. If the model invents a passage, the gate rejects it and forces a retry; if retries are exhausted, the answer is shown marked validation.ok = false, with lower confidence and a gap written to the audit: never presented as established fact. Returning a named INSUFFICIENT_EVIDENCE verdict instead of an uncited answer is our next step: the refusal mechanics exist, the verdict still needs exposing in the interface.

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6. Policy gaps and standards gaps

Each gap is a text we searched for and did not find, with the action it implies.

#	Type	Gap	Dim.	Recommendation
1	Law	notification duties not allocated across sectors	10, 7	designate a lead authority and an escalation path
2	Standard	no machine-readable traceability of automated decisions	5, 6	require an audit artefact (hash chain + citations) per decision
3	Policy	no guardrail on secondary use of inferred health signals	10, 9	prohibit advertising use; require consent and purpose limitation
4	Standard	health interoperability formats not binding	1, 3	make compliant formats and metadata mandatory
5	Policy	consent and purpose limitation for	6, 11	align with UNESCO and WHO

#	Type	Gap	Dim.	Recommendation
		minors' data unaddressed		guidance
6	Infrastructure	no redundancy requirement for critical e-health services	13	availability requirements for AI-assisted public services
7	Policy	no rule on liability when an official follows an AI recommendation and harm follows	10, 9	allocate liability, require a **named** approver, logged

Gap 7 is the one we can act on without waiting for the law: the approval flow is signed and nominative, so every decision carries a name.

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7. Architecture, Sovereignty, Auditability

SRC → C → PP → M → P → D → SINK (MLFO coordinator; the deployment is the ML sandbox)

One docker compose up, no hosted dependency: ollama (qwen3.5:9b, bge-m3), rerank (TEI, optional), qdrant, mongodb (audit, approvals, outbox), minio (content-addressed attachments and page images), ingestion, ocr (Tesseract), transcription (faster-whisper), mcp-charts and mcp-docx, api (Fastify, 81 routes), web (Next.js, 26 screens, 19 in-console, FR/EN). Twelve long-running services plus a one-shot MinIO bucket init.

- **Local by construction**: one inference mode, local; the console badge says so because the configuration accepts nothing else.
- **Sovereign auth**: scrypt passwords, TOTP with single-use recovery codes, local reset, sign-up closed by default; no identity provider.
- **E-mail without a provider**: notifications and approval links go to a console outbox and the operator log; no SMTP, no relay.
- **Untrusted context, verifiable record**: up to three images per question, content-addressed, treated as data never instructions; hash-chained log, persisted traces, every citation, exportable with chain verification.

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8. Evaluation-criteria alignment

Criterion	How the solution addresses it
Knowledge-base quality	26 entries / 21 ingestible, gazettes through local OCR with zone provenance, TITRE/CHAPITRE/ARTICLE tagging, ITU taxonomy, 10 regional strategies
Agent capability	mandated agents with partial information, citation-mandatory prompts, on-demand search, coordinator reconciliation, conflict detection
Gap identification	seven gaps across the 13 dimensions, each with a recommendation to the regulator
Auditability	deterministic citation gate, hash-chained log, traces, dossier export, chain re-verification, named approvals
ITU alignment	node-by-node Y.3172 mapping (MLFO, sandbox); Readiness 2.0 (6 factors, 13 dimensions); 0–5 scoring with causal what-if

Clickable evidence per dimension: **10 AI & Policies** (citations, rejected quotes, gap table) · **11 Inclusion** (bilingual console, local-only inference, OCR and voice) · **13 Infrastructure** (the self-hosted stack, offline once models are pulled) · **7 Strategy Alignment** (scoring against SND30 and PSNSN) · **6 Human Interface** (citation dialog, OCR overlay, signed approvals) · **5 Workflows** (ticks, approvals, notifications and audit in one flow).

Reproducibility. TypeScript/Python monorepo: 18 packages, 81 API routes, 26 screens. Validation for this submission: typecheck 38/38, test 228 tests in 44 files with 0 failure, production console build 20/20, docker compose config on all three stacks. Unit tests run without infrastructure; integration tests are environment-gated.

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9. Honest limits

- **Corpus coverage is partial.** Eight documents carry an OCR caveat (numbers flagged per document); five entries have no text at all (all to_confirm); six international references are catalogued but never loaded, so the system cannot quote Y.3172, WHO, UNESCO or the OECD principles.
- **The reasoning is a 9B local model**: bounded, slower than a hosted model, and its prose quality is not the claim; being gated, cited and logged is.
- **Not legal advice, not a production gate.** A real deployment needs the authority's own sign-off on the corpus.
- **The readiness scores are our model of judgement**, built on ITU dimensions and metrics, not an official ITU assessment.
- **No live integration with ministries**: inputs are the corpus, the scenario record, and what an operator uploads.

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10. From sandbox to deployment

Three concrete steps, in order: source the five to_confirm instruments and run OCR on the eight documents that need it, so the corpus claim is complete rather than honest-but-partial; expose the INSUFFICIENT_EVIDENCE verdict in the console and the dossier; and harden the approval path: mandatory reason, no self-approval, and an action type with no approval route at all, because some things are not ours to authorise. Each is a bounded change to code we already run, not a redesign.

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11. Team and submission

Role	Member
Team leader	Muller Roufaou
Members	[*names, max 4*]
Mentor	[*name if applicable*]

Repository: https://github.com/rouftom/sarkiai-policy-sandbox · Knowledge base: data/country/cm/ and data/global/ · Demo video: [*link*].

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